

A degree-free obstruction to linear runs of log-concavity failures in forests

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Abstract

Let $M(N)$ be the largest length of a consecutive run of strict log-concavity failures among independence polynomials of forests with at most N vertices, with no restriction on maximum degree. We prove the effective bound $M(N) < 8N/\log_2 \log_2 N$ for $N \geq 2^{65536}$. In particular $M(N) = o(N)$ uniformly over all forests, even when their maximum degree grows arbitrarily with their order. The key estimate is a local squared-correlation budget $\sum_{u \sim v} \text{Corr}(S_u, S_v)^2 \leq \lambda_v$ for the hard-core model with positive vertex activities. Weighted adjacency then controls short paths without counting graph-theoretic neighborhoods. Combining this with long-path decay, a convex-window variance identity, and entropy proves the bound. All thresholds are explicit; no asymptotic local limit theorem or numerical Fourier inversion is used.

Status and attribution. This note proves a universal upper bound, rather than another construction of exponent-one lower bounds. The latter are already reported for subcubic trees in Christopher’s September 2026 preprint [4]. The removal of every degree restriction is the substantive extension over our earlier bounded-degree ceiling. The proof below is self-contained. A finite exact-arithmetic audit accompanies it. Publication priority of this upper bound has *not* been established: the inspected primary sources do not state it, but Christopher’s full manuscript was inaccessible and its complete contents were not checked. This is a research draft, not an independently refereed or proof-assistant-verified result.

1 Statement with finite thresholds

For a forest F , let $i_k(F)$ count its independent sets of size k , and write $\alpha(F)$ for its independence number. A strict failure is

$$i_k(F)^2 < i_{k-1}(F)i_{k+1}(F), \quad 1 \leq k < \alpha(F).$$

A run consists of consecutive failure indices. Define

$$M(N) = \max_{\substack{F \text{ a forest} \\ |V(F)| \leq N}} \{\text{length of a strict failure run in } F\}.$$

The maximum is zero if no failure occurs. There is no degree parameter in this definition.

Theorem 1. *For each integer $m \geq 1$, put*

$$\Lambda_m = 4^m, \quad B_m = 2^{m+1}, \quad K_m = \lceil \log_2(12m^2) \rceil, \quad R_m = (\Lambda_m + 1)K_m. \quad (1)$$

Every forest of order $n \geq 24m^2 B_m^{R_m}$ has no strict failure run of length L with $L + 1 \geq n/m$. Consequently

$$M(N) < N/m \quad \text{for } N \geq N_m := 24m^3 B_m^{R_m}. \quad (2)$$

Corollary 2. For every integer $N \geq 2^{65536}$,

$$\boxed{M(N) < \frac{8N}{\log_2 \log_2 N}}. \quad (3)$$

In particular $M(N) = o(N)$.

Together with any valid subcubic lower bound $M_3(N) = N^{1-o(1)}$, this gives simultaneously

$$\lim_{N \rightarrow \infty} \frac{\log M(N)}{\log N} = 1, \quad \lim_{N \rightarrow \infty} \frac{M(N)}{N} = 0.$$

The second limit is now uniform over *all* forests. It does not follow by merely applying a theorem separately for each fixed degree. The exact subpolynomial factor remains undetermined.

2 A local correlation budget independent of degree

We first allow arbitrary positive vertex activities $\lambda_v \leq \Lambda$, where $\Lambda \geq 1$. The Gibbs weight of an independent set I is $\prod_{v \in I} \lambda_v$. Let $S_v = \mathbf{1}_{\{v \in I\}}$, $\mu_v = \mathbb{E} S_v$, $s_v^2 = \text{Var } S_v$, and $w_v = \mu_v / (1 - \mu_v)$. All s_v are strictly positive on a nonempty finite forest. For an oriented edge $u \rightarrow v$, let $R_{u \rightarrow v}$ be the root odds in the component containing u after deletion of that edge. Direct conditioning gives

$$R_{u \rightarrow v} = \lambda_u \prod_{z \sim u, z \neq v} (1 + R_{z \rightarrow u})^{-1}, \quad w_v = \lambda_v \prod_{u \sim v} (1 + R_{u \rightarrow v})^{-1}. \quad (4)$$

In particular $R_{u \rightarrow v} \leq \Lambda$.

Lemma 3. For every edge uv , let $a_{uv} = |\text{Corr}(S_u, S_v)|$. Then

$$a_{uv}^2 = w_u w_v, \quad \sum_{u \sim v} a_{uv}^2 \leq \lambda_v \leq \Lambda, \quad a_{uv} \leq q := \frac{\Lambda}{1 + \Lambda} < 1. \quad (5)$$

Proof. Adjacent vertices cannot both be occupied, so $\text{Cov}(S_u, S_v) = -\mu_u \mu_v$. Dividing its square by $s_u^2 s_v^2$ gives $a_{uv}^2 = w_u w_v$. Fix v and put $x_u = R_{u \rightarrow v}$. By the edge decomposition, $w_u = x_u / (1 + R_{v \rightarrow u}) \leq x_u$. Therefore

$$\sum_{u \sim v} a_{uv}^2 = w_v \sum_{u \sim v} w_u \leq \lambda_v \frac{\sum_{u \sim v} x_u}{\prod_{u \sim v} (1 + x_u)} \leq \lambda_v. \quad (6)$$

The final inequality uses $\prod (1 + x_u) \geq 1 + \sum x_u$ for nonnegative x_u . For the edge bound, set $R = R_{u \rightarrow v}$ and $S = R_{v \rightarrow u}$. Then $w_u = R / (1 + S)$ and $w_v = S / (1 + R)$, so

$$a_{uv}^2 = \frac{R}{1 + R} \frac{S}{1 + S} \leq q^2.$$

□

Let A be the symmetric weighted adjacency matrix with edge entries a_{uv} and all other entries zero. Root each component arbitrarily, and let $C_{vu} = a_{vu}$ if v is the parent of u , with zero entries otherwise. Rows of C have disjoint supports, so $CC^\top$ is diagonal and Lemma 3 gives $\|C\|_2 \leq \sqrt{\Lambda}$. Since $A = C + C^\top$,

$$\boxed{\|A\|_2 \leq 2\sqrt{\Lambda} =: B}. \quad (7)$$

This is an operator-norm bound, not a row-sum assertion: a large degree can make a row sum large. The proof uses the tree orientation and the squared-correlation budget instead.

3 Short paths, long paths, and a uniform variance bound

For vertices in one component, the tree Markov property yields the exact path identity

$$\text{Cov}(S_u, S_v) = (-1)^d s_u s_v \prod_{e \in P(u,v)} a_e, \quad d = \text{dist}(u, v). \quad (8)$$

One way to verify it is to use centered normalized variables $Z_v = (S_v - \mu_v)/s_v$. Since a centered function of a binary variable is a scalar multiple of that variable, on an edge $\mathbb{E}(Z_v \mid S_u) = \text{Corr}(S_u, S_v)Z_u$. Compose these identities along the path, using conditional independence of the separated subtrees. Different components have covariance zero.

Put $y_v = s_v$. For each integer $d \geq 0$, a pair at distance d has its unique simple path as one of the walks counted in $(A^d)_{uv}$. All these weights are nonnegative. Equations (8) and (7) therefore imply

$$\sum_{\text{dist}(u,v)=d} |\text{Cov}(S_u, S_v)| \leq y^\top A^d y \leq \|y\|_2^2 B^d \leq \frac{n}{4} B^d. \quad (9)$$

The distance-zero layer includes diagonal pairs. For distant pairs, (5) instead gives $|\text{Cov}(S_u, S_v)| \leq q^d/4$.

Lemma 4. *For any n -vertex forest with positive activities at most $\Lambda \geq 1$, any integer $R \geq 0$, and $X = \sum_v S_v$,*

$$\text{Var } X \leq \frac{2nB^R + n^2q^R}{4}, \quad B = 2\sqrt{\Lambda}, \quad q = \frac{\Lambda}{1 + \Lambda}. \quad (10)$$

Proof. Bound the variance by the sum of absolute covariances. Sum (9) for $0 \leq d \leq R$, using $\sum_{d=0}^R B^d \leq 2B^R$ because $B \geq 2$. There are at most n^2 remaining ordered pairs, each bounded by $q^R/4$, or with zero covariance. No count involving maximum degree is used. $\square$

For example, for every fixed Λ this proves a uniform polynomial saving from quadratic variance, including above spectral-independence thresholds. If $n \geq B^4$ and $R = \lfloor \log n / (2 \log B) \rfloor$, then

$$\text{Var } X \leq \frac{3}{4} n^{2-\eta}, \quad \eta = \frac{1}{4(1 + \Lambda) \log B} > 0. \quad (11)$$

Indeed $B^R \leq \sqrt{n}$, $R \geq \log n / (4 \log B)$, and $q^R \leq n^{-\eta}$ because $\log(1 + 1/\Lambda) \geq 1/(1 + \Lambda)$. Also $\eta < 1/2$, so $n^{3/2} \leq n^{2-\eta}$.

4 A convex coefficient window forces large variance

Suppose a failure run occupies indices $u, \dots, v$, with length $L = v - u + 1$. Put $d = L + 1$ and $c = (u + v)/2$. For the rest of the proof use one common activity $\lambda > 0$ and

$$p_k = \mathbb{P}_\lambda(X = k) = \frac{i_k \lambda^k}{Z_F(\lambda)}.$$

There is a unique activity for which $\mathbb{E}X = c$: the mean increases continuously from zero to $\alpha(F)$, with derivative $d\mathbb{E}X/d\log \lambda = \text{Var } X > 0$. The tilt preserves each strict reverse log-concavity inequality. Thus the probabilities are convex on $[u - 1, v + 1]$, because $p_{k-1} + p_{k+1} \geq 2\sqrt{p_{k-1}p_{k+1}} > 2p_k$.

For any sequence $z_0, \dots, z_d$, collecting coefficients proves

$$\begin{aligned} & \sum_{j=0}^d \left((j - d/2)^2 - \frac{d(d+2)}{12} \right) z_j \\ &= \sum_{k=1}^{d-1} \frac{k(k+1)(d-k)(d-k+1)}{12} (z_{k-1} - 2z_k + z_{k+1}). \end{aligned} \quad (12)$$

Apply this identity to $z_j = p_{u-1+j}$. The right side is nonnegative. For every integer outside this window, the squared distance from c is larger than $d(d+2)/12$. Adding the outside masses gives

$$\boxed{\text{Var}_\lambda X \geq \frac{d(d+2)}{12} > \frac{d^2}{12}.} \quad (13)$$

The verifier checks (12) as a polynomial identity in d, k , including its endpoint terms, rather than sampling window lengths.

The Gibbs entropy, in nats, is $H = \log Z_F(\lambda) - c \log \lambda$. Since $Z_F(\lambda) \geq \lambda^{\alpha(F)}$ and there are at most 2^n independent sets,

$$(\alpha(F) - c) \log \lambda \leq H \leq n \log 2.$$

The enlarged window ends at $v+1 \leq \alpha(F)$, so $\alpha(F) - c \geq d/2$, and therefore

$$\boxed{\lambda \leq 2^{2n/d}.} \quad (14)$$

These bounds hold at the actual finite-order centered activity.

5 Proof of the universal ceiling

Suppose $d \geq n/m$. Equation (14) gives $\lambda \leq 4^m = \Lambda_m$. Set the radius to R_m from (1). Since $(1 + 1/\Lambda_m)^{\Lambda_m+1} > 2$ by the binomial theorem,

$$\left(\frac{\Lambda_m}{1 + \Lambda_m} \right)^{R_m} < 2^{-K_m} \leq \frac{1}{12m^2}.$$

If $n \geq 24m^2 B_m^{R_m}$, Lemma 4 implies

$$\text{Var } X \leq \frac{n^2}{48m^2} + \frac{n^2}{48m^2} = \frac{n^2}{24m^2} < \frac{d^2}{12},$$

contradicting (13). For the maximum over all orders up to $N \geq N_m$, a forest of order $n < 24m^2 B_m^{R_m}$ has every run shorter than N/m trivially. For larger n , the proved bound is $L < n/m \leq N/m$. This proves Theorem 1. $\square$

An explicit envelope. For every integer $m \geq 1$,

$$K_m \leq 4m, \quad m \leq 2^m, \quad m+1 \leq 2^m, \quad R_m \leq 8m4^m \leq 2^{3m+3}.$$

For the first inequality, $12m^2 \leq 16m$ follows by induction: it holds at one and $16m^2 \geq (m+1)^2$ for $m \geq 1$. The next two bounds also follow by induction from $m=1$. Consequently

$$\log_2 N_m < 5 + 3m + (m+1)R_m \leq 5 + 3m + 2^{4m+3} \leq 2^{4m+4}.$$

The last inequality uses $5+3m \leq 2^{4m+3}$, true at one and preserved when the right side is multiplied by 16. Thus $N_m \leq 2^{4m+4}$. For $N \geq 2^{65536} = 2^{2^{16}}$, put $A = \log_2 \log_2 N \geq 16$ and $m = \lfloor (A-4)/4 \rfloor$. Then $4m+4 \leq A$ and $m \geq (A-8)/4 \geq A/8$. Apply (2) to obtain (3). $\square$

6 A degree-free polynomial barrier in the bulk

Corollary 5. Fix $0 < \delta < 1$ and put

$$\varepsilon_\delta = \frac{1}{8(1+2^{2/\delta})(1+1/\delta)\log 2} > 0.$$

If a failure run in an n -vertex forest has midpoint $c \leq (1-\delta)\alpha(F)$ and $n \geq 2^{4+4/\delta}$, then

$$\boxed{L+1 < 3n^{1-\varepsilon_\delta}.} \quad (15)$$

No maximum-degree assumption is required.

Proof. A forest is bipartite, so $\alpha(F) \geq n/2$. At the centered activity the entropy argument yields $\lambda \leq \Lambda := 2^{2/\delta}$. Now $B = 2\sqrt{\Lambda} = 2^{1+1/\delta}$, and the assumed threshold is exactly $n \geq B^4$. Apply (11) with $\eta = 2\varepsilon_\delta$ and combine it with (13). $\square$

Thus in any sequence of forests, even of unbounded degree, runs of length $n^{1-o(1)}$ must have midpoint $c/\alpha(F) \rightarrow 1$. Otherwise Corollary 5 gives a fixed polynomial saving on an infinite subsequence. Since $L = o(n)$ by Corollary 2 and $\alpha(F) \geq n/2$, both endpoints of such runs also have ratio tending to one with $\alpha(F)$. This is a statement about the location of *near-linear runs*; it does not exclude isolated failures or shorter runs in the bulk.

7 Verification and relation to existing results

Run the standard-library-only program

```
python verify_degree_free.py
```

It writes `degree_free_certificate.json`. The executed run passed 8119 checks: 16 universal algebra/constant checks, threshold checks, and an exact finite audit on 25 graph/activity cases containing 2230 ordered covariance pairs. The cases include paths, stars, binary trees, a star of stars, and a disconnected forest, at both uniform and nonuniform rational activities. The audit independently enumerates independent sets and checks marginal message identities, the squared-correlation budget, signed path factorization, weighted-walk bounds and the variance estimate. It avoids numerical square roots by using the rational matrix similar to A , with edge entries $\mu_v/(1 - \mu_u)$. The symbolic checks prove algebraic identities for indeterminates; the graph enumeration is a finite sanity audit, not an exhaustive proof. The universal probability and operator arguments are Sections 2–5.

Degree-independent spectral estimates for trees are established tools: Efthymiou, Hayes, Štefankovič, and Vigoda [1] prove spectral independence for activities below 1.3, and Chen, Yang, Yin, and Zhang [2] extend a threshold to e^2 . Our bound (11) is much weaker than a uniform spectral-independence bound in those regimes, but it applies at every finite activity ceiling, which is essential because $\Lambda_m = 4^m$ grows without bound. Fang et al. [3] prove eventual unimodality of forests and log-concavity on a central interval, using activities in $[1/4, 12]$. The present argument controls arbitrary reverse-log-concavity windows, including ones outside that interval, without extending their local limit theorem. Christopher’s indexed abstract [4] reports near-linear lower constructions and an exact covariance threshold. We could not retrieve its full proof, so no claim that this upper bound is absent from that work is made. Priority remains an explicit unresolved issue.

References

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- [3] E. X. Fang, J. Lu, E. Nevo, Y. Yao, H. Zheng, *Unimodality of Independence Polynomials for Sufficiently Large Forests*, arXiv:2609.20961v1, 17 September 2026. <https://arxiv.org/abs/2609.20961>.
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